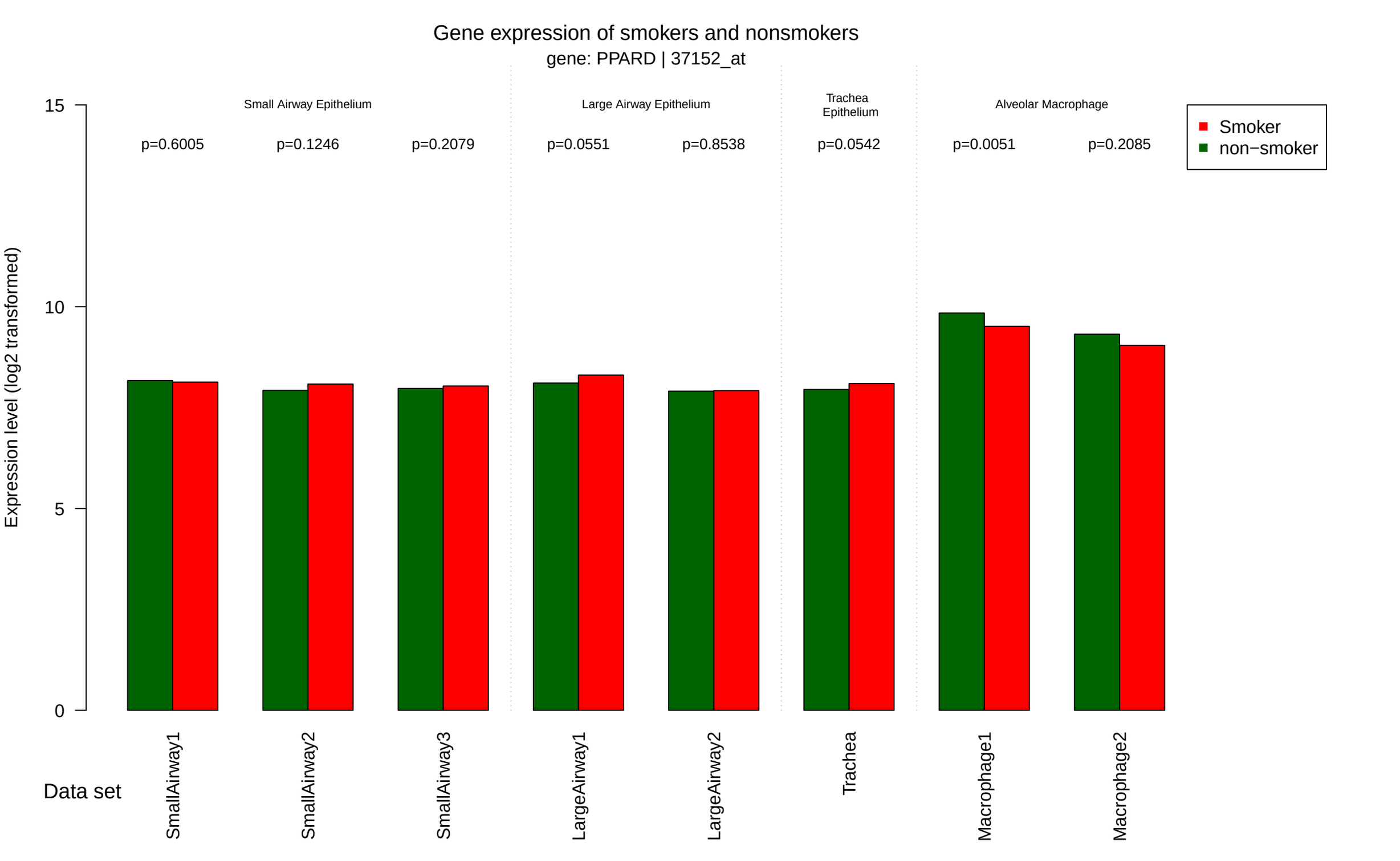
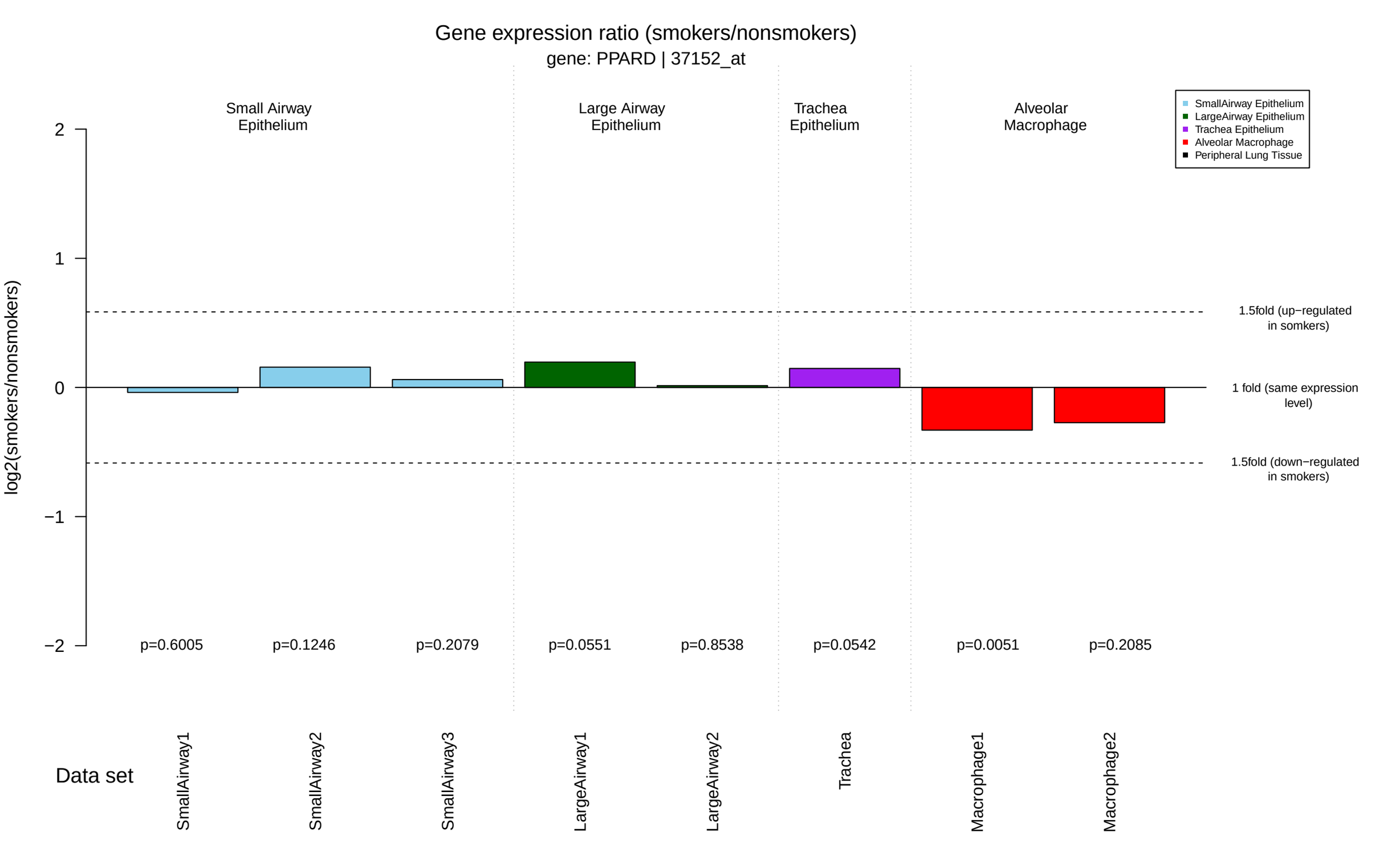
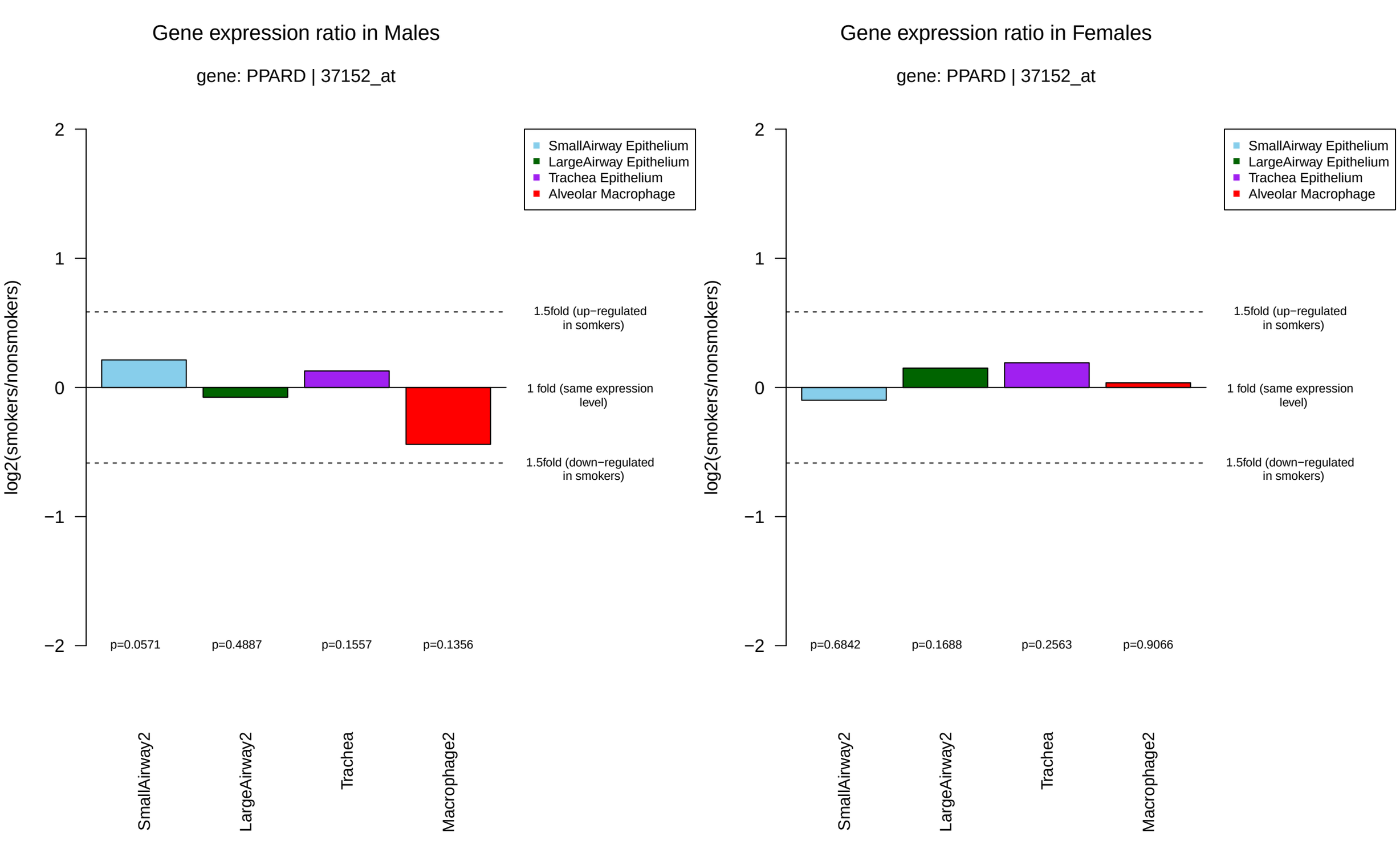
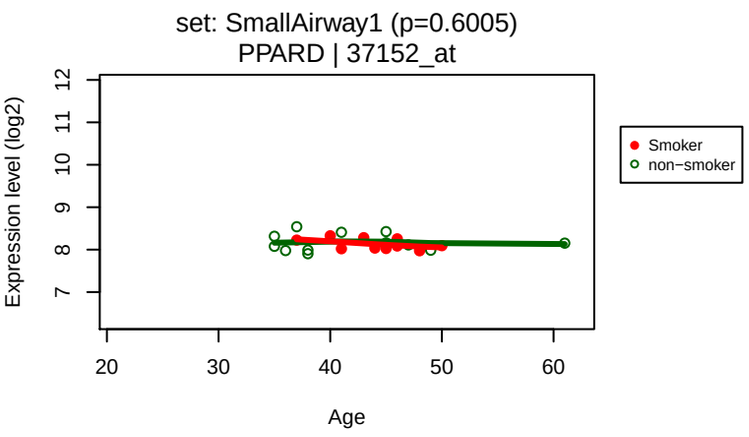
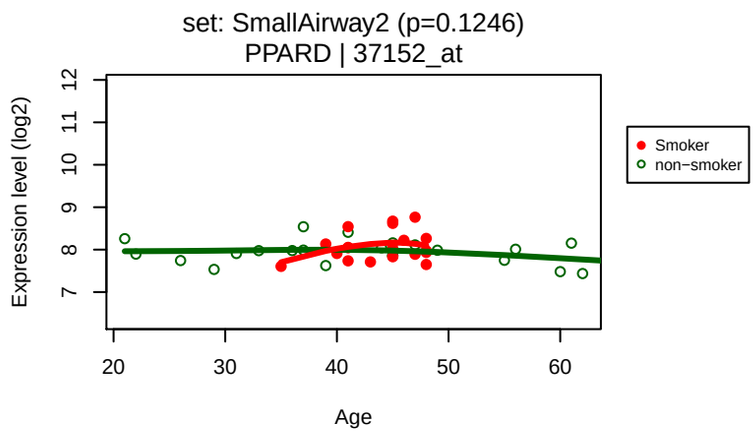
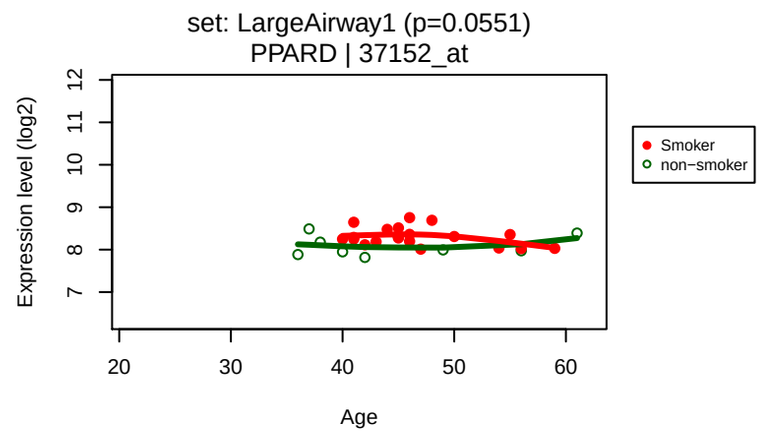

Gene expression of smokers and nonsmokers
gene: PPARD | 37152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6005
p=0.1246
p=0.2079
p=0.0551
p=0.8538
p=0.0542
p=0.0051
p=0.2085
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

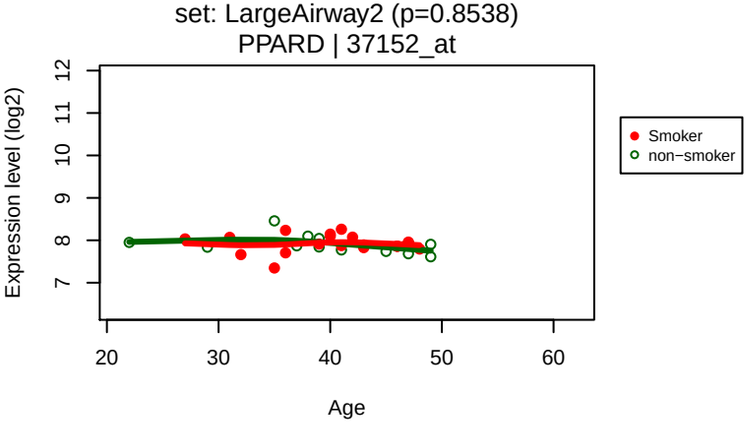

Gene expression ratio (smokers/nonsmokers)
gene: PPARD | 37152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6005
p=0.1246
p=0.2079
p=0.0551
p=0.8538
p=0.0542
p=0.0051
p=0.2085
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

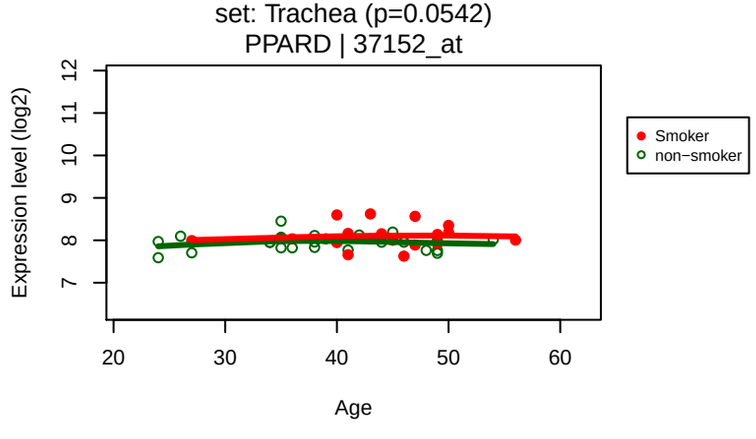

Gene expression ratio in Males
gene: PPARD | 37152_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0571
p=0.4887
p=0.1557
p=0.1356
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARD | 37152_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6842
p=0.1688
p=0.2563
p=0.9066
SmallAirway2
LargeAirway2
Trachea
Macrophage2

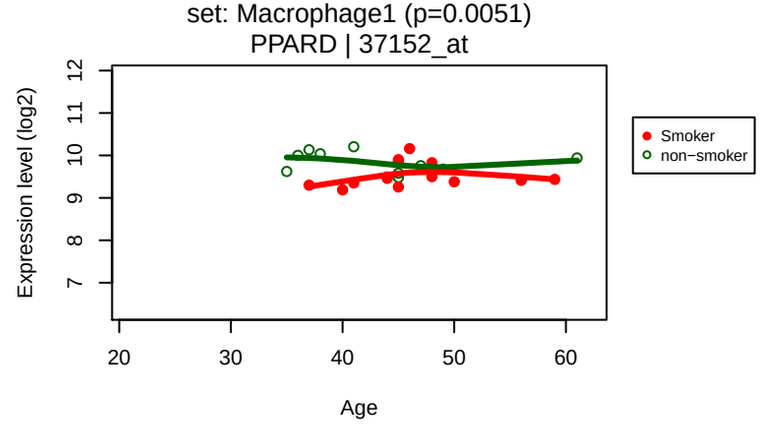

set: SmallAirway1 (p=0.6005)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker

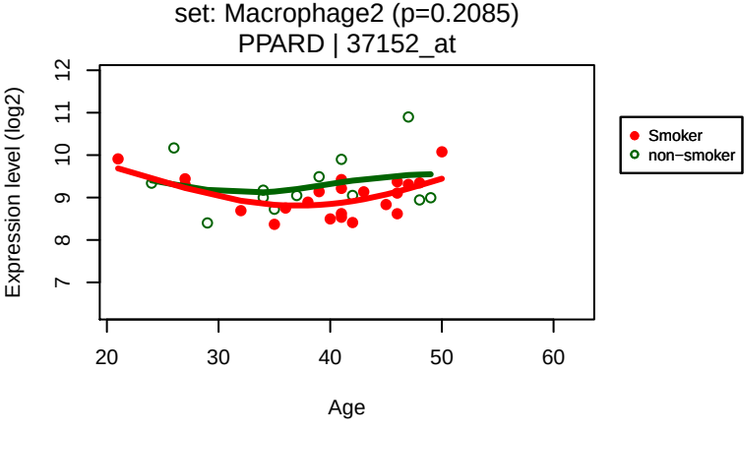

set: SmallAirway2 (p=0.1246)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker

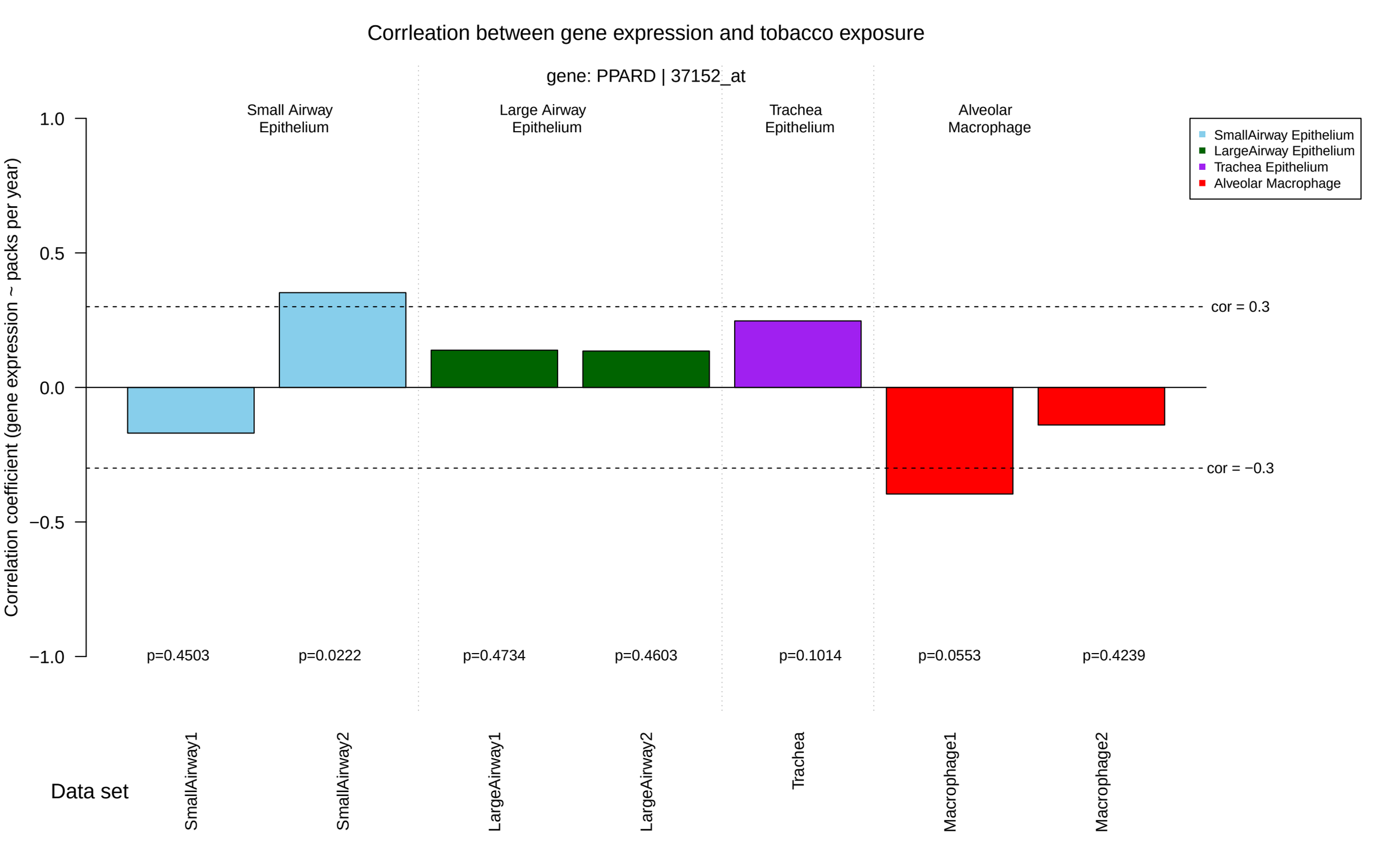

set: LargeAirway1 (p=0.0551)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8538)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0542)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0051)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2085)
PPARD | 37152_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARD | 37152_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4503
p=0.0222
p=0.4734
p=0.4603
p=0.1014
p=0.0553
p=0.4239
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2